

Lecture 10: System Bus (Data Bus, Address Bus & Control Bus)

Learning Objectives

By the end of this lecture, students will be able to:

- Define a System Bus.
 - Explain why buses are needed in a computer.
 - Differentiate between Data Bus, Address Bus, and Control Bus.
 - Understand bus width and its impact on performance.
 - Explain memory read and memory write operations.
 - Understand CPU–Memory communication.
 - Relate buses to Assembly Language and computer architecture.
-

1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally include:

"In today's Computer Organization and Assembly Language lecture, we'll learn the System Bus, including the Data Bus, Address Bus, Control Bus, bus width, memory read and write operations, and how the CPU communicates with memory using simple diagrams and examples."

In our previous lectures, we explored the internal components of the CPU, including the Arithmetic Logic Unit (ALU), Control Unit (CU), and CPU Registers.

Now here's an important question:

How does the CPU communicate with RAM?

How does it receive data from a keyboard?

How does it send information to a monitor or printer?

The answer is through a communication system called the **System Bus**.

Just as roads connect different cities, buses connect the CPU, memory, and input/output devices so they can exchange information.

In today's lecture, we'll learn about the three major buses used in computer systems: the **Data Bus**, **Address Bus**, and **Control Bus**. We'll also understand how memory read and write operations work.

Let's begin.

2. What is a System Bus?

Definition

A System Bus is a collection of communication pathways that transfer data, addresses, and control signals between the CPU, memory, and input/output devices.

In simple words:

A bus is a communication channel that allows computer components to exchange information.

Without buses:

- The CPU cannot access RAM.
 - The CPU cannot receive keyboard input.
 - The CPU cannot display output on the monitor.
-

3. Why Do We Need a Bus?

Imagine every department in a university has to communicate.

Without roads, people cannot travel.

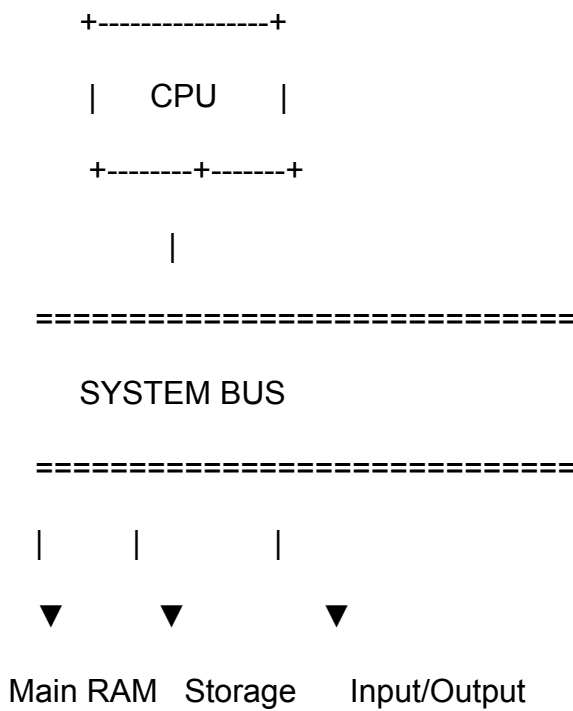
Similarly:

- CPU
- RAM
- SSD
- Keyboard
- Monitor

all need a communication path.

That communication path is called the **System Bus**.

4. Overall Bus Structure



Everything communicates through the system bus.

5. Types of Buses

There are three major buses.

System Bus

|

|— Data Bus

|— Address Bus

└— Control Bus

Each has a different purpose.

6. Data Bus

Definition

The Data Bus transfers actual data between the CPU, memory, and input/output devices.

Examples of data:

- Numbers

- Characters
- Images
- Instructions
- Audio

Example

Suppose RAM contains:

25

CPU requests it.

The value:

25

travels through the **Data Bus**.

Characteristics of Data Bus

- Transfers data.
- Usually **bidirectional**.
- Connects CPU, memory, and I/O devices.

Diagram:

CPU

↔

Data Bus

↔

RAM

Data can move in both directions.

7. Address Bus

Definition

The Address Bus carries the memory address that the CPU wants to access.

It does **not** carry the actual data.

Example:

Suppose the CPU needs data stored at memory location:

1024

The Address Bus carries:

Address = 1024

Memory then locates that address.

Characteristics of Address Bus

- Carries addresses only.
- Usually **unidirectional**.

- Normally flows from CPU to memory or I/O.

Diagram:

CPU



Address Bus



Memory

8. Control Bus

Definition

The Control Bus carries control and timing signals that coordinate operations between the CPU and other components.

Common control signals include:

- Read
- Write
- Interrupt
- Reset
- Clock
- Memory Enable
- I/O Enable

Example

Suppose the CPU wants to read memory.

The Control Bus sends:

READ

If the CPU wants to store data:

WRITE

Characteristics

- Transfers control signals.
- Many signals travel from the CPU to other devices, while some (such as interrupts) travel back to the CPU.
- Coordinates hardware operations.

Diagram:

CPU

↔

Control Bus

↔

9. Summary of the Three Buses

Bus	Carries	Direction (Typical)
Data Bus	Data	Bidirectional
Address Bus	Addresses	CPU → Memory/I/O
Control Bus	Control Signals	Both directions (depending on signal)

A simple way to remember:

- **Address Bus → Where?**
 - **Data Bus → What?**
 - **Control Bus → How and When?**
-

10. Bus Width

A bus contains multiple electrical lines.

The number of lines is called the **Bus Width**.

Examples:

- 8-bit

- 16-bit
 - 32-bit
 - 64-bit
-

Example

An 8-bit Data Bus:

10110110

Transfers **8 bits** at a time.

A 32-bit Data Bus transfers **32 bits** in one transfer.

Generally, a wider data bus can move more data per transfer.

11. Address Bus Width

The width of the Address Bus determines how many unique memory locations can be addressed.

Example:

8-bit Address Bus

$2^8 = 256$ addresses

16-bit Address Bus

$2^{16} = 65,536$ addresses

32-bit Address Bus

$2^{32} = 4,294,967,296$ addresses

The general formula is:

Number of addresses = 2^n

where n = Address Bus width

12. Memory Read Operation

Suppose the CPU wants to read the value at address **1000**.

Step 1

The CPU places **1000** on the Address Bus.

↓

Step 2

The Control Bus sends the **READ** signal.

↓

Step 3

Memory finds address **1000**.

↓

Step 4

The stored value travels to the CPU on the Data Bus.

Diagram:

CPU



Address Bus



Memory



Data Bus



CPU

13. Memory Write Operation

Suppose the CPU wants to store the value **50** at address **1000**.

Step 1

The CPU places **1000** on the Address Bus.



Step 2

The CPU places **50** on the Data Bus.



Step 3

The Control Bus sends the **WRITE** signal.



Step 4

Memory stores **50** at address **1000**.

Diagram:

CPU



Address = 1000



Data = 50



WRITE Signal



Memory

14. Complete Communication Example

Suppose:

`int x = 25;`

Execution:

CPU



Address Bus



Memory

↓

Data Bus

↓

CPU

↓

Registers

↓

ALU

The buses transfer the information needed for the CPU to execute the instruction.

15. Real-Life Analogy

Imagine ordering a book from a library.

- **Shelf Number** → Address Bus
- **Book** → Data Bus
- **"Please bring this book"** → Control Bus

The shelf number tells *where* the book is.

The book itself is the data.

The request to bring it is the control signal.

16. Bus Performance

Bus performance depends on:

- Bus Width
- Clock Frequency
- Bus Architecture
- System Design

A wider or faster bus can generally transfer more information over time.

17. Relationship with CPU Registers

Suppose the CPU executes:

```
int z = x + y;
```

Process:

RAM

↓

Data Bus



Registers



ALU



Registers



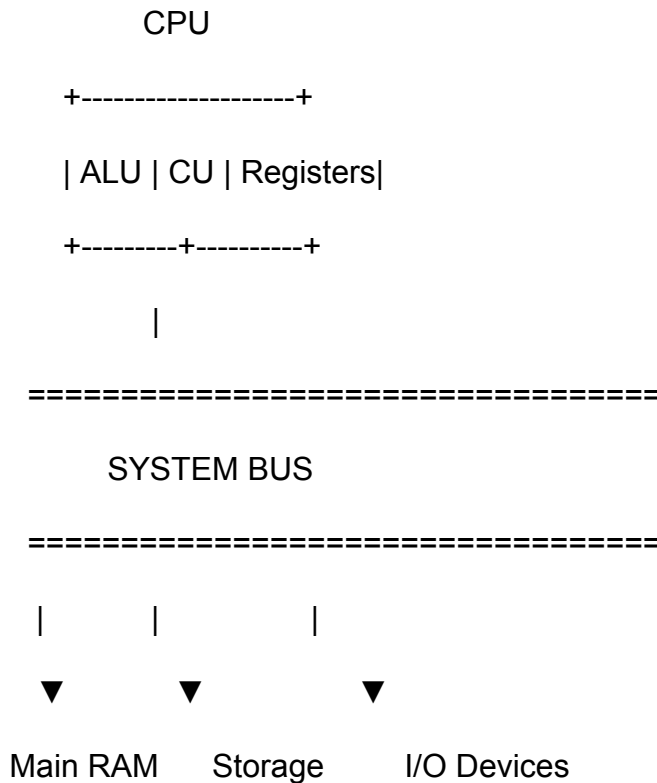
Data Bus



RAM

The buses carry the data, while the registers temporarily hold it.

18. System Bus Diagram



Lecture Summary

Today, we studied the **System Bus**, the communication pathway that connects the CPU with memory and input/output devices.

We learned about the three major buses: the **Data Bus**, which transfers actual data; the **Address Bus**, which specifies the memory location to access; and the **Control Bus**, which carries control and timing signals.

We also explored bus width, memory read and write operations, and how these buses work together to allow the CPU to communicate with the rest of the computer.

In the next lecture, we'll study one of the most important topics in Computer Organization—the **Fetch–Decode–Execute Cycle**. We'll see exactly how the CPU

executes every instruction step by step using the Program Counter, Instruction Register, MAR, MDR, the Control Unit, the ALU, and the System Bus.

Thank you for watching, and I'll see you in the next lecture.
